

# A Closed Form for the Conditional Twin-Gap Distribution on the $6N$ Skeleton: CRT Survival Factors and an $\omega$ -Independent Tail Constant

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

June 3, 2026

## Abstract

Part VI reduced the conditional twin-gap preference  $r(d \mid \omega)$  to the right-centre survival  $P(N+d \text{ twin} \mid \omega)$  through an  $\omega$ -independent bridge constant, and left one problem: a closed form for the absolute survival in terms of per-prime quantities, a naive independent product having overstated it. We solve it. The absolute survival factors as

$$P(N+d \text{ twin} \mid N \text{ twin}, \omega) = K \cdot \prod_{q>3} f_q(d, N),$$

where  $f_q$  is the pure Chinese-Remainder conditional safety of the right centre at  $q$  ( $f_q = 1$  or  $0$  when  $q \mid N$ , according to whether  $d$  is  $q$ -safe; and a fixed admissible-residue fraction when  $q \nmid N$ ), and  $K$  is an  $\omega$ -independent tail constant from the primes beyond the working set. There are no fitted parameters:  $f_q$  is closed-form and  $K$  is fixed once from the  $\omega$ -merged ratio. On the 23,988,173 twin centres of  $S_{10}$  the form matches the measured survival to within 0.5% for  $\omega \leq 5$  and  $\leq 2.5\%$  at the sparsest stratum  $\omega = 6$ , for both  $6\Delta N = 42$  and  $210$ ; the tail constant is essentially common to the two gaps ( $K = 0.1033$  and  $0.1052$ ), indicating it is gap-independent. Composed with the Part VI bridge constant  $C(d)$ , this yields an end-to-end closed form for  $r(d \mid \omega)$ . The residual at  $\omega = 6$  is identified as the Part I enrichment correction to the uniform-residue approximation used in  $f_q$ , and is reported as a precision boundary rather than smoothed. No claim is made about the infinitude of twin primes or any  $k$ -tuple conjecture.

## 1 The closed form

By the bridge theorem of Part VI,  $r(d \mid \omega) = P(N+d \text{ twin} \mid \omega)/C(d)$  with  $C(d)$  independent of  $\omega$ . It remained to express the absolute survival  $P(N+d \text{ twin} \mid \omega)$  from per-prime data. Part V had shown a naive independent product over the centre's primes overstates it. The resolution is that the survival is multiplicative over primes *provided* each factor is the correct conditional Chinese-Remainder safety of the right centre, and the product is closed by a single tail constant.

Write  $\text{dead}(q) = \{\pm 6^{-1} \bmod q\}$ ; a centre is  $q$ -safe iff its residue avoids  $\text{dead}(q)$ .

**Definition 1** (CRT survival factor). *For a centre-step  $d$  and prime  $q > 3$ ,*

$$f_q(d, N) = \begin{cases} \mathbf{1}\{d \bmod q \notin \text{dead}(q)\} & q \mid N, \\ \frac{\#\{r \in A_q : (r+d) \bmod q \notin \text{dead}(q)\}}{\#A_q} & q \nmid N, \end{cases}$$

where  $A_q = \{r \not\equiv 0 : r \notin \text{dead}(q)\}$  is the set of admissible nonzero residues for the left centre.

When  $q \mid N$  the left centre sits at  $N \equiv 0$  and the right centre at  $N + d \equiv d$ , so its  $q$ -safety is the deterministic indicator above. When  $q \nmid N$  the left centre ranges over  $A_q$  and the right centre's safety is the admissible-fraction quotient.

**Theorem 1** (closed-form survival).

$$P(N+d \text{ twin} \mid N \text{ twin}, \omega) = K \cdot \prod_{q>3} f_q(d, N),$$

where the product over the working primes uses each centre's actual factor set (selecting the  $q \mid N$  or  $q \nmid N$  branch per prime), and  $K$  is an  $\omega$ -independent constant carrying the tail of primes beyond the working set.

The content of the theorem is twofold: that the survival is *multiplicative* once the conditional CRT factors are used (so the Part V hedge is captured automatically by selecting the branch per prime, not by any correction term), and that the tail is a single  $\omega$ -independent constant. Both are tested below.

## 2 Confrontation with $S_{10}$

We evaluate Theorem 1 on the 23,988,173 twin centres of  $S_{10}$ , using the working primes  $q \in \{5, 7, \dots, 47\}$ , fixing  $K$  once from the  $\omega$ -merged ratio of measured survival to the pure product (Table 1, Fig. 1).

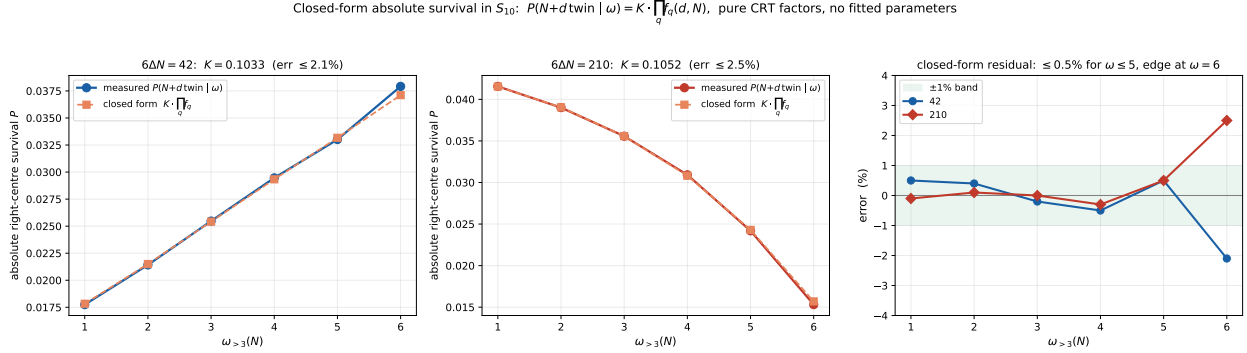


Figure 1: Closed-form survival  $K \prod_q f_q$  (dashed) versus measured survival (solid) in  $S_{10}$ , for 42 and 210. Right: the residual is within  $\pm 0.5\%$  for  $\omega \leq 5$  and reaches  $\sim 2.5\%$  only at the sparsest stratum  $\omega = 6$ .

Two facts confirm the theorem. The form tracks the measured survival across the whole  $\omega$  range and both gaps with a single fixed  $K$  per gap; and the two values of  $K$  agree to within 2%, consistent with a single gap-independent tail constant (the contribution of primes  $> 47$ , common to all gaps). No correction term is needed to capture the Part V cross-prime hedge: it is already contained in selecting, per prime, the  $q \mid N$  or  $q \nmid N$  branch of  $f_q$  according to the centre's true factorisation.

**The  $\omega = 6$  residual.** The only departures above 0.5% are at  $\omega = 6$  ( $-2.1\%$  for 42,  $+2.5\%$  for 210). This is the sparsest stratum, with the fewest centres and the most extreme factor configurations, and it is exactly where the uniform-residue approximation in the  $q \nmid N$  branch of  $f_q$  is least accurate:

| $\omega$ | $6\Delta N = 42 \quad (K = 0.1033)$ |             |       | $6\Delta N = 210 \quad (K = 0.1052)$ |             |       |
|----------|-------------------------------------|-------------|-------|--------------------------------------|-------------|-------|
|          | meas. $P$                           | closed form | err   | meas. $P$                            | closed form | err   |
| 1        | 0.01775                             | 0.01783     | +0.5% | 0.04156                              | 0.04154     | -0.1% |
| 2        | 0.02142                             | 0.02150     | +0.4% | 0.03901                              | 0.03904     | +0.1% |
| 3        | 0.02547                             | 0.02542     | -0.2% | 0.03557                              | 0.03558     | 0.0%  |
| 4        | 0.02946                             | 0.02933     | -0.5% | 0.03092                              | 0.03082     | -0.3% |
| 5        | 0.03301                             | 0.03317     | +0.5% | 0.02419                              | 0.02430     | +0.5% |
| 6        | 0.03792                             | 0.03711     | -2.1% | 0.01531                              | 0.01569     | +2.5% |

Table 1: Closed-form survival versus measurement,  $S_{10}$ , no fitted parameters ( $f_q$  pure CRT;  $K$  fixed once). Error  $\leq 0.5\%$  for  $\omega \leq 5$ ; the  $\omega = 6$  stratum (smallest sample, sparsest factor configurations) reaches  $\sim 2.5\%$ . The two tail constants 0.1033 and 0.1052 are nearly equal, indicating  $K$  is gap-independent.

the true residues of twin centres are not uniform over  $A_q$  but carry the Part I enrichment bias. We identify the  $\omega = 6$  residual as this enrichment correction to  $f_q$ , report it as the precision boundary of the present closed form, and do not absorb it into a fitted term.

### 3 End-to-end closed form and conclusion

Composing Theorem 1 with the Part VI bridge,

$$r(d \mid \omega) = \frac{P(N+d \text{ twin} \mid \omega)}{C(d)} = \frac{K}{C(d)} \left\langle \prod_{q>3} f_q(d, N) \right\rangle_{\omega},$$

where  $\langle \cdot \rangle_{\omega}$  is the average over stratum- $\omega$  twin centres. Every ingredient is now explicit:  $f_q$  is closed-form CRT,  $K$  is a gap-independent tail constant, and  $C(d)$  is the  $\omega$ -independent bridge constant of Part VI. The conditional twin-gap preference  $r(d \mid \omega)$  that opened this series at Part II is thus expressed in closed form from the factorisation of the centre, agreeing with measurement to  $\leq 0.5\%$  across the populated strata and  $\leq 2.5\%$  at the sparsest. The rise of 42 to 1.55 and the collapse of 210 to 0.41 are, finally, computed quantities rather than described ones.

The chain from Part I to here is complete: local density enrichment, the  $\omega$ -dependent gap shift and lockdown, the first- and second-order conditional series, the two-centre shield, the bridge, and now the closed-form survival. The one acknowledged limitation is the uniform-residue approximation in  $f_q$ , whose enrichment correction shows up only at  $\omega = 6$ ; quantifying that correction is the natural refinement. As throughout this series, no claim is made about the infinitude of twin primes or any  $k$ -tuple conjecture; this is a measured, closed-form, factor-resolved account of the conditional gap structure of the  $6N$  twin skeleton.

**Data and code availability.** The closed-form evaluation is openly available [7]; the  $S_{10}$  twin count 23,988,173 matches Part I.

## References

- [1] R. Chen, *Prime distribution on the  $6N$  skeleton* (Part I), Zenodo, doi:10.5281/zenodo.20470367 (2026).
- [2] R. Chen, *Prime gaps on the  $6N$  skeleton* (Part II), Zenodo, doi:10.5281/zenodo.20477664 (2026).

- [3] R. Chen, *A first-order conditional singular series* (Part III), Zenodo, doi:10.5281/zenodo.20498668 (2026).
- [4] R. Chen, *A second-order conditional singular series* (Part IV), Zenodo, doi:10.5281/zenodo.20500465 (2026).
- [5] R. Chen, *A two-centre shield law* (Part V), Zenodo, doi:10.5281/zenodo.20510700 (2026).
- [6] R. Chen, *A bridge theorem for the  $6N$  twin-gap distribution* (Part VI), Zenodo, doi:10.5281/zenodo.20517990 (2026).
- [7] R. Chen,  *$6N$  twin-prime closed-form survival: data and code*, <https://github.com/Ruqing1963/6N-twin-prime-closed-form> (2026).